

Take Home Project

Senior Data Scientist

Introduction

Congratulations! This take-home project gives you the opportunity to showcase your real-world skillset. We encourage the use of AI tools, but we will ask you detailed questions about your implementation, so make sure you have reviewed and can explain everything that AI has done. In this project, you can choose the coding languages that allow you to showcase your best work. At Natural State, the Tech Team (software developers) work in Python and SQL while the Biometrics Team (scientists) work in R. Our platform, Natural State Analytics, has a tech stack containing: Python 3.13, R, PostgreSQL, Redis, React 18+Typescript+Vite5, ODK, Megadetector+SpeciesNET, BirdNet. The platform infrastructure is: Azure, Terraform, Docker Compose, Github, Jira.

Time limit

We expect this project to take 4-6 hours. We will not monitor this, we trust you to limit yourself.

Deliverables

Provide your submission as Markdown documents. These should include text, code, graphs, dashboards, or any other artifacts you feel are appropriate. Don't just share outputs, explain what you did, why you did it, and the decisions you made along the way. There is no single right answer, so we value clear communication of your reasoning and tradeoffs.

Please also create a new public GitHub repository for this assignment and use it throughout your development process. We encourage you to commit your work incrementally so we can review not only the final solution, but also how your thinking and implementation evolved over time. We are not looking for a specific number of commits or a particular Git workflow. Meaningful and natural progress updates are preferred over artificially splitting work into many small commits. The goal is simply to help us understand your approach, decision-making process, and development journey.

Questions

We encourage you to ask questions before you begin by email.

Background

Natural State works with different biodiversity restoration projects. We help projects design their biodiversity data collection forms using Open Data Kit (ODK). These forms are completed by the project

team in the field using the ODK app, and the forms are then automatically uploaded to our Natural State Analytics (NS Analytics) platform, managed by the Tech team. Once the forms reach the platform, they need to be go through a cleaning, joining, and QA/QC process on the back end before they are used to create dashboards and reports for the project team.

Projects collect different types of field data using different methods. For example, vegetation data (herbaceous and woody) is collected using ODK to record human field observations (e.g., vegetation transects). Bird data is collected using passive acoustic monitoring devices with deployment metadata tracked using ODK. SD cards used in passive acoustic monitoring devices are uploaded to the Natural State Analytics platform using our desktop uploader app, which links every SD card with the relevant deployment metadata from ODK at the time of upload.

Bird Challenge

Once all the data is on the Natural State Analytics platform, audio recordings are run through the BirdNET machine learning classifier. BirdNET segments audio recordings and assigns a confidence score to each segment as to whether or not one or more labeled classes (i.e., species) are vocalizing in that segment. A prediction is any instance where a confidence score exceeds a user-specified minimum value. Each segment with a prediction is stored, but there are too many segments for every prediction to be validated by a human ornithologist. The challenge is to determine the threshold confidence score for each class above which predictions constitute observations.

Natural State follows an approach outlined by Wood and Kahl ([2024](#)) to evaluate the relationship between confidence scores and true positive predictions to determine which predictions, if any, exceed a species-specific threshold associated with very high certainty (e.g., precision ≥ 0.99) and can be classified as observation. For each species, Natural State uses the ornithologist-validated clip segments to plot “logit transformed confidence scores” on the x-axis and “pr(BirdNET prediction is correct)” on the y-axis. Using the fitted curve, Natural State sets confidence score thresholds for each species, above which predictions can be interpreted as observations (e.g. when BirdNET predicts “Olive Thrush” with a 0.56 confidence score we can be 99% sure that that prediction is correct). Using this logic, all clips can be labeled with their bird species observation.

1. We have provided you with a sample bird dataset containing BirdNET predictions and human ornithologist validations for a subset of BirdNET predictions. Using a 99% probability of correctness cutoff, assign each clip in the ‘birdnet_predictions’ file a label in a new “observation” field.
2. What information would you send to the Tech team such that the approach you used can be implemented on the NS Analytics platform and run automatically once projects upload their audio recordings? Assume that the SD card upload to ML pipeline is already functional, and all that’s

missing is what comes after the ML predictions are recorded. Be sure to include an explanation of how you derived observations from predictions, and what decisions you made along the way.

Vegetation Challenge

In the field, teams collect data on herbaceous vegetation by surveying quadrats along a transect line. They follow the attached SOP to do this, and record data in ODK. The purpose of the data is to assess herbaceous species richness and diversity.

We have provided you with the ODK form used to capture herbaceous vegetation survey data, the corresponding entity lists, and the raw data outputs. The data needs to be cleaned, joined, and run through a QA/QC process. Importantly, you should not change or fix any data errors you find – we only need mistakes to be flagged, not resolved.

1. Produce a “data quality report” for the project. The sections of the report that you need to populate are:
 - a. Survey-level summary
 - b. Plot-level summary
 - c. Assessment of sampling effort
 - d. Map of transect locations
2. What information would you send to the tech team such that the approach you used can be implemented on the NS Analytics platform to run automatically and produce a Vegetation QA/QC dashboard for the project? Assume that the data collected by ODK has already reached the platform, all that’s missing is what comes after the raw data tables arrive.

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